

Q3. The Perfect Number

A positive integer is called a **perfect number** if it happens to equal to the sum of all its factors (not including itself). For example, say 28, the sum of all its factors is $1+2+4+7+14=28$. Therefore, 28 is called a perfect number. For 24, the sum of all its factors is $1+2+3+4+6+8+12=36$, which is not equal to 24. As a result, 24 is not a perfect number. Write a program to determine a number N (For simplicity, assume $N < 10000$) is a perfect number or not.

Input:

A number N

Output:

If N is a perfect number, print “N is a perfect number”. Otherwise, print “N is not a perfect number” instead.

Input example:

28

24

0

-8

6

Output Example:

28 is a perfect number

24 is not a perfect number

0 is not a perfect number

-8 is not a perfect number

6 is a perfect number